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Major crustal rotations in the Andean margin: Paleomagnetic results from the Coastal Cordillera of northern Chile

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Abstract. Paleomagnetic analyses of Mesozoic lavas and dike swarms from the northern Chilean Coastal Cordillera, between 25.4°S and 26.4°S, reveal a clockwise rotation of about 42°. Magnetizations from lava flows of andesitic-basaltic composition of the Middle Jurassic La Negra Formation pass both fold and reversal tests and are interpreted as prefolding remanences. Five dike swarms of Middle Jurassic to Early Cretaceous age yield similar directions to that obtained from the La Negra Formation. Four of the five swarms have mixed polarity, suggesting that they too carry a primary or very early remanence. The structural setting of the dikes suggests that they have not suffered any substantial tilting about nonvertical axes since acquisition of the remanence. The clockwise rotation of the area is believed to have been the consequence of transpressional deformation of mid-Late Cretaceous age, post-100 Ma, associated with abandonment of the Jurassic-Early Cretaceous magmatic arc in this region and its eastward migration to form a new mid-Late Cretaceous magmatic arc in the former back arc region. This younger arc is located east of the Coastal Cordillera and lies in the Central Valley region. The clockwise sense of rotation is consistent with other paleomagnetic data from northern Chile and southern Bolivia, south of the Arica Deflection in the Andean margin, although it is the largest yet reported. To the north of the Arica Deflection, paleomagnetic studies report counterclockwise rotations, and several large-scale models have been proposed to explain the overall pattern of rotations. Models include oroclinal bending of an originally straight margin, differential shortening across the margin at a preexisting bend which is subsequently tightened by the passive rotation of the limbs of the bend, and distributed shear throughout the margin as a consequence of oblique convergence at a preexisting bend. In contrast to these models, several workers have argued that rotation is better explained in terms of localized in situ rotations. We review these models in light of our results and present a domino-type model with blocks bounded by left-lateral faults and rotating clockwise in response to mid-Late Cretaceous transpression within a crustal scale shear zone. This is consistent with the observed strike-slip fault systems identified in the Coastal Cordillera.

Introduction

A major feature of the Andean margin is the ~55° change in strike of both topography and geology at ~19°S which is commonly referred to as the Arica Deflection or Bolivian Orocline. Paleomagnetic studies in this part of the Andean margin of northern Chile, southern Peru, and Bolivia have identified substantial vertical axes rotations such that the general pattern of paleomagnetic rotations is counterclockwise north of the bend and clockwise to the south (Figure 1 and Table 1). Several different hypotheses have been advanced to explain the orogenic deflection and the subsequently recognized paleomagnetic rotations, including those reported from the Mesozoic magmatic arc of the Coastal Cordillera of northern Chile. These hypotheses include (1) oroclinal bending, that is, the essentially rigid bending of an originally straight margin around a fixed hinge zone [e.g., *Carey*, 1955; *Kono et al.*, 1985]; (2) differential shortening across a preexisting bend [e.g., *Isacks*, 1988; *Roperch and Carlier*, 1992; *MacFadden et al.*, 1995]; (3) oblique convergence at a preexisting bend [*Beck*, 1987, 1988, and (4) various forms of local rotations about near vertical axes [e.g., *Hartley et al.*, 1988; *Forsythe and Chisholm*, 1994; *Somoza*, 1994]. Crustal rotation is therefore recognized as an important expression of deformation in the Andean margin; however, no consensus has yet been reached concerning the type of mechanism or mechanisms responsible for these rotations nor the horizontal dimensions of the crustal blocks involved. We believe that the diversity of models partly results from the lack of detailed geological control, in particular, the limited kinematic study of fault systems in the areas selected for paleomagnetic study.

The overall aim of our study has been to investigate the deformation of the overriding plate at a subduction margin, and in particular to investigate the contributions of extension, compression, translation, and rotation of crustal blocks through time in relation to the fault systems observed. Our approach has been to integrate structural, geochronologic, and paleomagnetic in-

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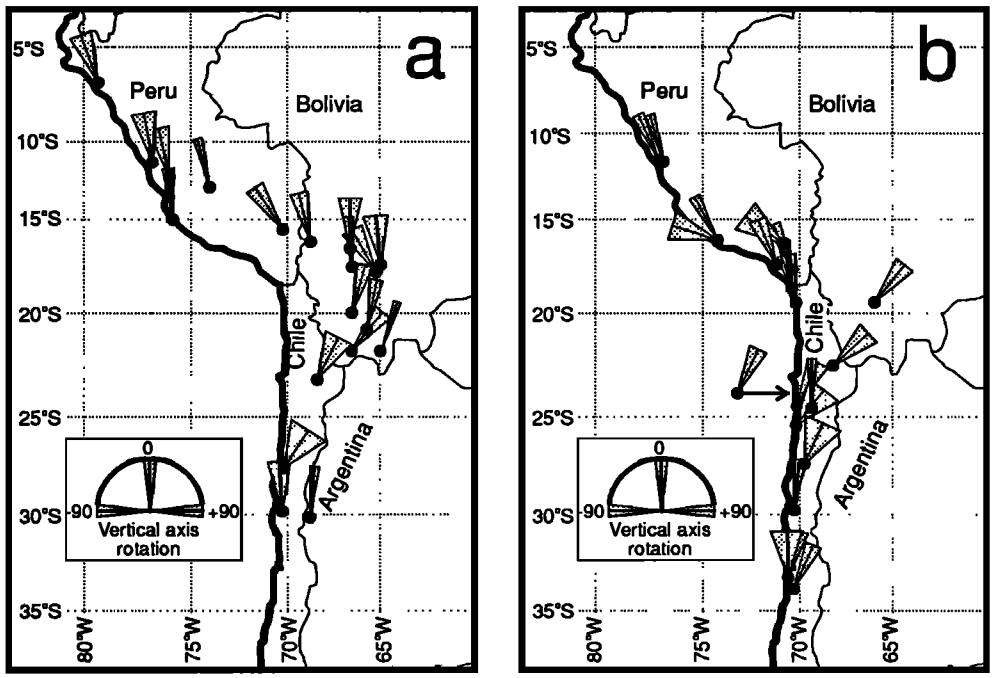


Figure 1. Paleomagnetic rotations identified in the Andean margin in the region of the Arica Deflection; (a) results from Tertiary units, (b) results from pre-Tertiary units, as in Table 1.

formation to investigate and distinguish deformational episodes and their geological signatures. In particular, the principal structural feature of the Coastal Cordillera of northern Chile is the Atacama Fault Zone which is believed to be traceable for some 1100 km, from Iquique in the north (20°S) to La Serena in the south (30°S) [*Naranjo,* 1987]. This important fault zone is known to have had a major left-lateral strike-slip component in Mesozoic times and is thought to be a trench linked strike-slip fault zone [*Sylvester,* 1988; *Grocott et al.,* 1994a] which decouples the forearc sliver (or terrane) from the main continental plate. It was therefore thought likely to have played an important role in forearc block rotations and to have the potential to be the locus for substantial latitudinal transport of the outboard forearc sliver.

The region under investigation extends from approximately 25.4°S to 27.4°S and lies west of 70°W (Figure 2). In this paper we present the results of our paleomagnetic investigations from an area between 25.4°S - 26.4°S and, mainly, from west of the major strands of the Atacama Fault Zone (AFZ). The principal objective of this study has been to provide a paleomagnetic data set to evaluate the magnitude of rotation and to identify the contribution, if any, of slip along the AFZ to the observed rotations. Paleomagnetic results from the Upper Triassic Cifuncho Formation in the north of the detailed study area had already indicated that at least part of the area had suffered substantial rotation [*Forsythe et al.,* 1987], but the age of remanence acquisition and the regional extent of this rotation and its timing were uncertain.

Several previous paleomagnetic studies in the Coastal Cordillera of northern Chile have suffered from restricted areal coverage and/or insufficient sampling to account adequately for paleosecular variation. They have suffered from poor determinations of the age of the sampled rocks, the age of remanence acquisition, and the timing of fault activity within the area. We have attempted to address these problems by extensively sampling rocks of Early Jurassic to Early Cretaceous age over a substantial area (100 x 30 km.). The area is dominated by the ductile-brittle N-S trending AFZ and a pervasive NW set of brittle faults (Figure 3), the detailed kinematics of which have been the focus of structural investigations [*Brown et al.,* 1993; *Grocott et al.,* 1994a,b]. The chronology of pluton and dike emplacement and ductile fault displacements has been tightly defined by high-precision $^{40}Ar/^{39}Ar$ geochronology [*Dallmeyer et al.*, 1996]. Integration of structural, geochronologic, and paleomagnetic data with the regional remapping of the area by the geological survey of Chile (SERNAGEOMIN) leads us to believe that we have identified the faults responsible for rotation and geological interpretation of the age of displacement on these faults allows us to define the age of rotation. We present a model of block rotation within a left-laterally driven crustal scale shear zone compatible with the observed deformation of the region.

Geological Setting

The basement of the study area comprises upper Paleozoic metasedimentary rocks [*Bell,* 1987] intruded by a broad belt of genetically related Permian to Early Triassic granitic plutons (Figure 2) [*Brown,* 1991]. Inboard of this basement, and separated from it by a major extensional fault system, the Mesozoic magmatic arc is defined by plutons of Late Triassic to Early Cretaceous age. The Middle Jurassic to Early Cretaceous plutons are notably elongate in outline with their long axis parallel to the magmatic arc and inferred subduction margin. The plutons, their associated mylonitic wall rocks, brittle fault systems, and the sedimentary and volcanic sequences all show a systematic eastward younging within the arc. Some plutons have been dated using both $^{40}Ar/^{39}Ar$ [*Dallmeyer et al.,* 1996], and U-Pb methods [*Berg and Breitkreuz,* 1983] and yield identical ages within error. Geobarometry using the Al-in-hornblende

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Table 1. Paleomagnetic Rotations Reported From Close to the Arica Deflection in the Andean Margin

Locality	Latitude, °S	Longitude, °W	Age, Ma	Rotation (R±ΔR), deg.	Reference
			Tertiary		
North of Arica Deflection (<17°S)					
Cajamarca	6-7	79.0	Early Oligocene	-26.6±4.9	*Mitouard et al.* [1990]
Volcanics	11-15	76-77	Eocene-Miocene	-18.6±4.2	*Macedo-Sánchez et al.* [1992a]
Volcanics	11-13	76-78	Eocene	-16.5±7.9	*Macedo-Sánchez et al.* [1992b]
Ocros	13.0	74.0	8.5	-14.2±5.0	*Heki et al.* [1983, 1985]
Sediments and volcanics	14.7	76.1	Miocene-Pliocene	-5.3±6.7	*Tsunakawa et al.* [1987]
Laguna Umayo	15.8	70.1	~60	-43.9±9.6	*Butler et al.* [1995]
Punta Japones-Chama	16-17	68.5	Oligocene-Miocene	-14.0±9.7	*Roperch et al.* [1993]
Within hinge zone (17°-19°S)					
Salla	17.0	67.0	28-24	-6.6±5.4	*MacFadden et al.* [1990]
Micaña	17.5	67.0	7.5-6.5	-4.8±5.2	*MacFadden et al.* [1990]
Tiupampa	18.0	65.5	60-58	-12.8±8.0	*Butler et al.* [1995]
Sucusuma	18.1	65.8	61-59	-35.4±5.6	*Butler et al.* [1995]
South of Arica Deflection (>19°S)					
Quehua	20.0	67.0	13-7	15.3±9.5	*MacFadden et al.* [1995]
Cerdas	21.0	66.0	16.5-15.5	10.1±7.2	*MacFadden et al.* [1995]
Lipez	22.0	67.0	20	39.0±11.0	*Roperch et al.* [1993]
Quebrada Honda	22.0	65.0	13-12	17.8±4.1	*MacFadden et al.* [1990]
Paciencia Group	22.6	68.3	~28	20.0±12.2	*Hartley et al.* [1992a]
Elquinos Formation	29.5	70.5	?Early Cenozoic	-4.0±13.0	*Palmer et al.* [1980]
Río Azul sediments	30.6	68.9	~12.5-9.2	5.3±5.4	*Jordan et al.* [1990]
Cordillera de Domeyko	27.5	70.1	?Cenozoic	26.1±21.2	*Riley et al.* [1993]
			Pre-Tertiary		
North of Arica Deflection (<17°S)					
Puente Piedra Formation	11.9	77.1	~90	-19.9±4.3	*May and Butler* [1985]
Sediments/volcanics	12.0	76.9	~112-78	-28.2±5.6	*Macedo-Sánchez et al.* [1992b]
Chala	15.8	74.3	~170	-68.5±15.9	*Roperch and Carlier* [1992]
Chala	15.8	74.3	~80	-38.2±4.9	*Roperch and Carlier* [1992]
Aréquipa	16.5	71.8	~100	-60.5±18.6	*Roperch and Carlier* [1992]
Within hinge zone (17°-19°S)					
Ilo	17.5	71.4	~80	-32.2±9.4	*Roperch and Carlier* [1992]
La Yarada	18.1	70.7	~80	-32.5±6.8	*Roperch and Carlier* [1992]
Arica Dikes	18.6	70.3	Early Cretaceous	-18.0±4.7	*Heki et al.* [1985]
South of Arica Deflection (>19°S)					
Camaraca/Cuya Dikes	19.2	70.3	Middle Jurassic-Early Cretaceous	-7.7±8.7	*Scanlan and Turner* [1992]
La Palca	19.5	65.9	71-60	32.9±4.7	*Butler et al.* [1995]
Purilactis Formation	22.8	68.4	Late Cretaceous-Early Cenozoic	41.0±8.8	*Hartley et al.* [1992a]
Coastal Cordillera	24.0	70.0	Middle Jurassic-Early Cretaceous	28.3±8.8	*Hartley et al.* [1988]
Cordillera de Domeyko	24.5	69.0	Triassic[a]	0±5.7	*Jesinkey et al.* [1987]
Coastal Cordillera	24.5	70.5	Jurassic-Cretaceous	28.8±11.6	*Forsythe and Chisholm* [1994]
Cifuncho Formation	25.6	70.6	?Early Cretaceous	37.2±21.1	*Forsythe et al.* [1987]
Cordillera de Domeyko	27.5	69.5	mid-Cretaceous	20.6±22.3	*Riley et al.* [1993]
Volcanics	29.5	70.5	Early-mid Cretaceous	6.3±6.7	*Palmer et al.* [1980]
Coastal Cordillera	33.0	71.5	Jurassic	3.0±15.6	*Irwin et al.* [1987]
Coastal Cordillera	33.0	71.5	Jurassic	12.4±14.1	*Irwin et al.* [1987]
Central Valley	34.0	70.7	Late Cretaceous	13.7±8.1	*Beck et al.* [1986]

The rotations are calculated using the parameters of *Beck* [1980] and *Demarest* [1983].
[a] As no Triassic reference is calculated in Beck [1988], the original quoted rotation is used.

method indicates shallow emplacement depths which coupled with the geochronology and field relations indicates that the Middle Jurassic to Early Cretaceous plutons were emplaced at high crustal levels and cooled rapidly [*Dallmeyer et al.*, 1996]. During the Jurassic and Early Cretaceous, the plutons were emplaced at jogs in an extensional fault system which was linked to the incipient Atacama Fault Zone [*Grocott et al.*, 1994a]. The tectonic setting for this period is that of a retreating subduction margin in which the overall rate of convergence is lower than the rate of subduction and deformation in the overriding plate is characterized by extension [*Royden*, 1993].

The Late Triassic to Early Cretaceous plutons are intruded by dense dike swarms whose spatial and close temporal relationships to their host pluton suggest that they represent a final phase of emplacement of the pluton itself [*Dallmeyer et al.*, 1996]. For example, dikes which intrude the Flamenco and Las Animas plutons both gave dates of circa 154 Ma, close to the age of the Las Animas pluton itself. One dike from the Las Tazas pluton dike swarm yields a date of circa 129 Ma, which is within the spread of ages (132-127 Ma) defined from the host pluton [*Dallmeyer et al.*, 1996, also unpublished data, 1996]. As the dikes were to be sampled for paleomagnetic purposes,

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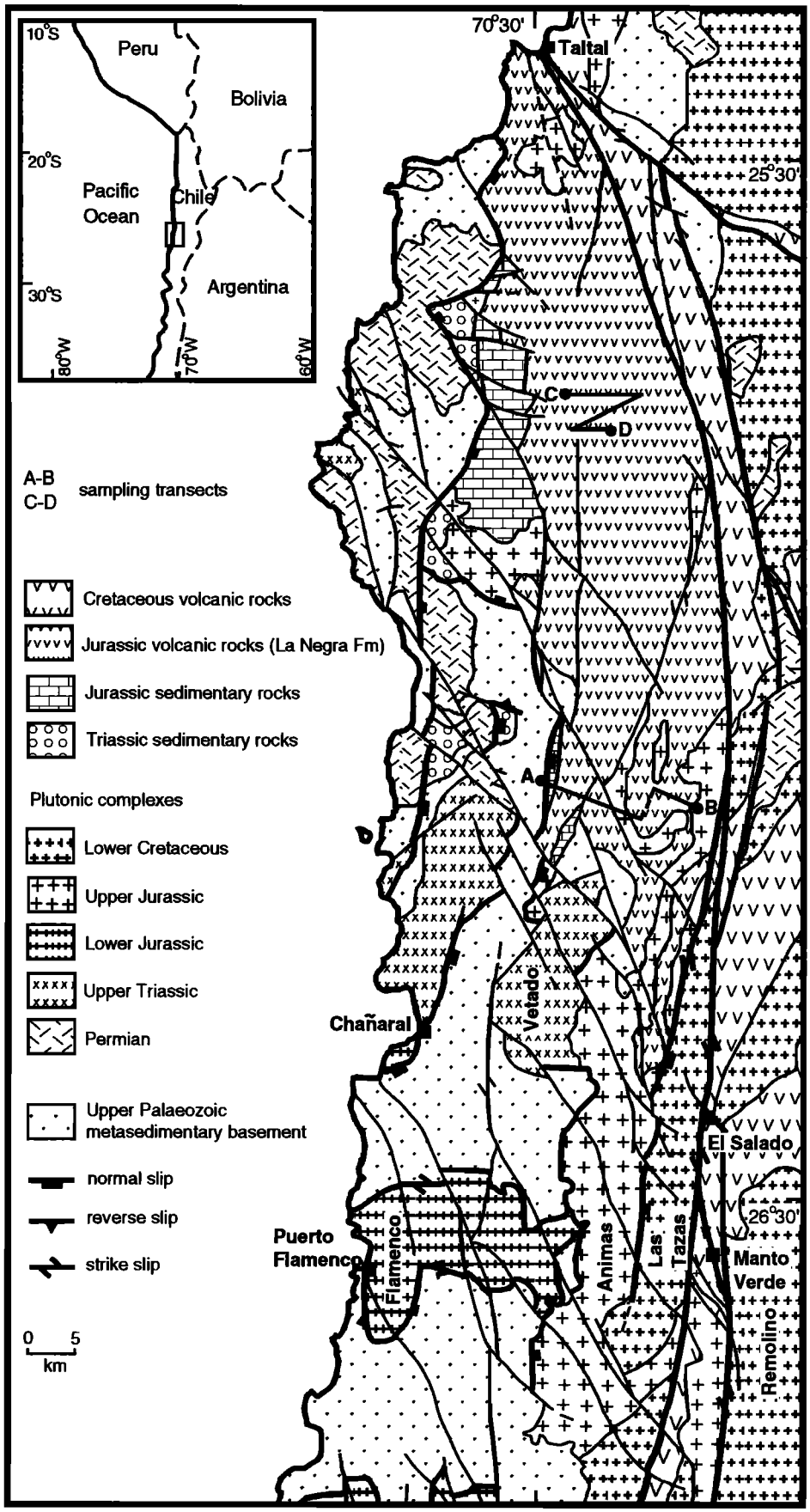


Figure 2. Geologic map of the detailed study area between 24.4°S and 26.4°S with the general locations of the paleomagnetic sampling sites.

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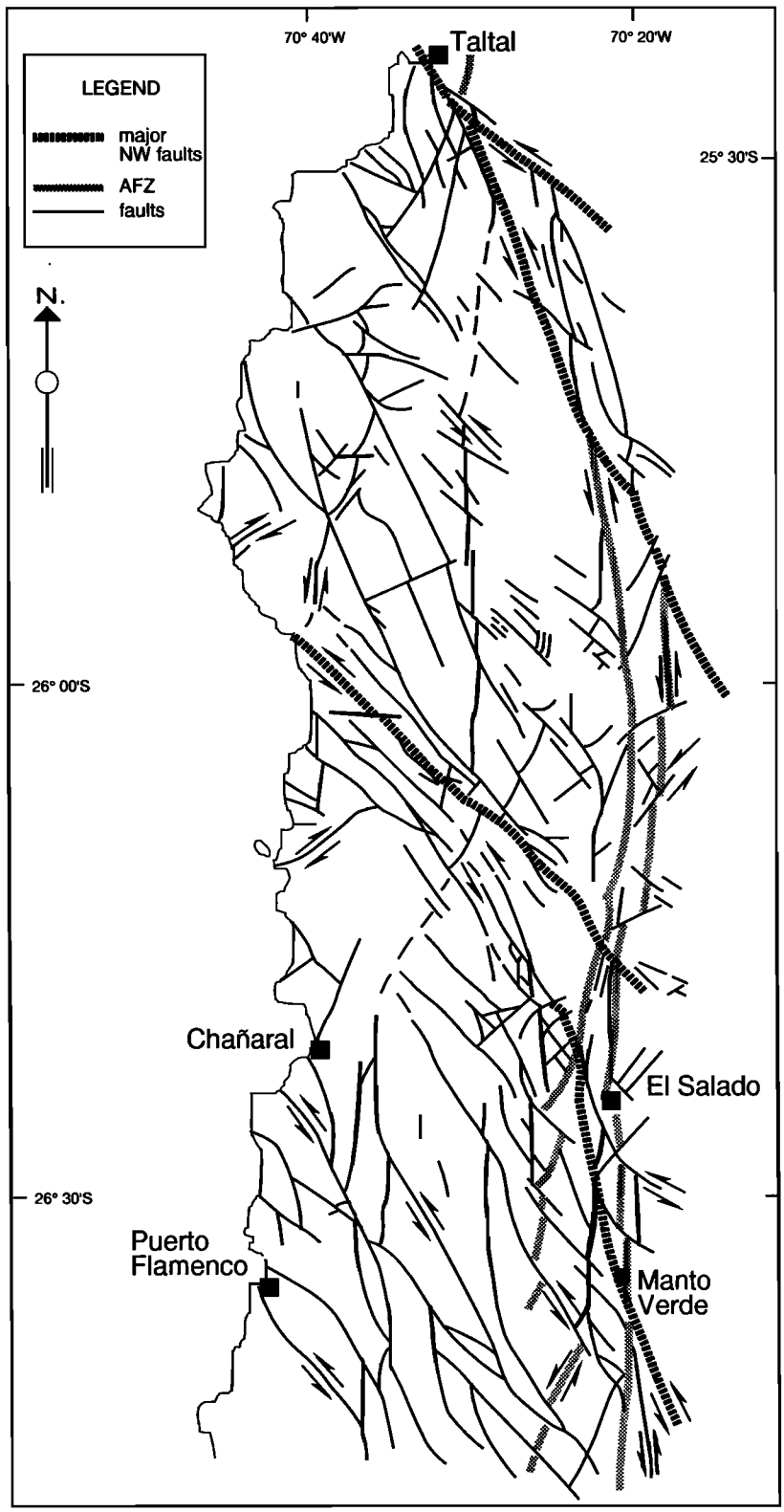


Figure 3. Fault map for the detailed study area compiled from our own field observations and image interpretation. The principal strands of the AFZ and major crosscutting NW trending faults are identified.

we have investigated their structural setting. Some 200 measurements of the dilation directions for three dike swarms from the Late Triassic Vetado, Middle Jurassic Las Animas, and Early Cretaceous Las Tazas plutons were taken. Stereonet analysis of the dilation planes [*Bussell*, 1989] reveals that all three swarms have horizontal maximum extension directions, although the azimuth of these extension directions varies significantly. Given the difference in the timing of the intrusion of these swarms and the fact that all yield horizontal extension directions, it would be improbable that they have suffered tilting,

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between or after emplacement, which could result in such a unique geometrical situation. We interpret the data therefore to indicate that the maximum extension direction was close to horizontal during emplacement for all the swarms, as might reasonably be expected and that none of them have suffered significant postemplacement tilting.

The sedimentary and volcanic sequences of the north of the area include the Cifuncho (Upper Triassic), Pan de Azúcar (Lower Jurassic), La Negra (Lower-Middle Jurassic) and Bandurrias (Lower Cretaceous) Formations (Figure 2). These formations are separated from the basement to the west by a major extensional fault system which was active as a growth fault during the Triassic as suggested by the rapid changes in facies and thickness of the Cifuncho Formation [*Suarez and Bell*, 1992]. The La Negra Formation is widely exposed throughout the Coastal Cordillera and has been previously sampled for paleomagnetic studies [*Hartley et al.*, 1988; *Tanaka et al.*, 1988]. It is early Sinemurian to Kimmeridgean in age, based on stratigraphic and paleontologic evidence [*Garcia*, 1967; *Naranjo*, 1978] and comprises lavas with intercalated volcaniclastic sediments and breccias. The depositional setting, in this area, is shallow marine in the west to subaerial in the east [*Naranjo*, 1978], however, as the formation dips mainly eastward this may be a temporal change due to younging representing emergence, rather than a strictly spatial change. The La Negra and underlying Pan de Azúcar Formation are affected by tilting of fault-bounded blocks in an extensional fault system which is the equivalent of the ductile faults controlling contemporary pluton emplacement in the south of the area (Figure 2). These formations are also deformed by NW-SE trending, low-amplitude, very shallowly plunging open folds. To date, these folds have only been identified to the north and west of the main paleomagnetic sampling areas, although their more widespread presence is suspected.

The Coastal Cordillera is cut by the major Atacama Fault Zone (AFZ) which is broadly subdivided into three concave west segments which, from north to south, are known as the Salar del Carmen, Paposo, and El Salado segments [*Naranjo*, 1987]. Our study area is located in the northern part of the El Salado segment of the AFZ which is here defined by a series of ductile and brittle, subparallel and en echelon north-south curvilinear faults (Figure 3). The AFZ shows evidence of only very minor right-lateral and dip-slip motion [*St. Amand and Allen*, 1960; *Arabasz*, 1971; *Hervé*, 1987] which is a result of post Miocene reactivation during uplift of the Coastal Cordillera. During the Jurassic and Cretaceous, both the ductile and later brittle strike-slip phases of movement are consistently and dominantly left-lateral in at least the Paposo and El Salado segments [*Hervé*, 1987; *Scheuber and Andriessen*, 1990; *Brown et al.*, 1993]. The principal ductile left-lateral movements in this segment of the fault are dated as Early Cretaceous in age (circa 132-126 Ma) [*Dallmeyer et al.*, 1996, also unpublished data, 1996]. The ductile and brittle strike-slip movements are part of a single deformational event with heat for the ductile deformational phase being supplied the plutons themselves [*Grocott et al.*, 1994a].

A major feature of the region is a set of large, steeply dipping, NW-SE trending, brittle, left-lateral oblique-slip faults some of which cut and displace the AFZ, while others sole into the AFZ (Figure 3). Most of these were initiated in response to the strike-slip displacement on the AFZ but were later extended and rejuvenated such that the largest of these faults displace the main strands of the AFZ. The largest identified displacement on these NW faults is some 70 km [*Brown et al.*, 1993], which we interpret as separating the original, clearly ductile El Salado and Paposo segments of the AFZ. These NW trending faults can be traced to the eastern margin of the Coastal Cordillera [*Arevalo*, 1994, 1995], where they sole out into a N-S to NE-SW fault zone. This fault zone we refer to as the Central Valley Shear Zone, which divides the Coastal Cordillera from the Central Valley which are distinct physiographic and geological domains. The major NW trending left-lateral strike-slip faults and the Central Valley Shear Zone we collectively term the Atacama Fault System (AFS).

The youngest pluton of the detailed study area, which is exposed in the extreme south, is dated at 106 Ma [*Dallmeyer et al.*, 1996] and is part of the final phase of magmatic activity in this region. About this time (circa 100 Ma) the locus of magmatic activity migrated some 50 km eastward into what had previously been the back arc basin. This new mid-Late Cretaceous magmatic arc developed within what is the present-day Central Valley [*Scheuber and Reutter*, 1992], east of the Central Valley Shear Zone. Arc abandonment in the Coastal Cordillera and eastward migration correlates with the first major phase of Andean compression, the Mirano Orogeny, which is remarkably synchronous along the entire Andean margin and is interpreted to be a consequence of a major and rapid change in plate convergence [*Dalziel*, 1983]. In the studied area this is expressed as a change from transtensional strike-slip tectonic on the AFZ to transpressional strike-slip tectonics on the AFS. The NW trending faults of the AFS in the Copiapo area are postdated by the deposition of the Cerillos Formation which is Aptian to Albian in age [*Arevalo*, 1995] and provides a minimum age constraint for this fault activity contemporaneous with arc abandonment. This marked change in tectonic and magmatic activity we interpret as the consequence of the margin becoming an advancing type of subduction boundary in which the overall convergence rate is greater than the rate of subduction and deformation in the overriding plate is characterized by contraction [*Royden*, 1993]. Subsequent to this, the Coastal Cordillera records no significant tectonic activity until the Miocene when it suffered uplift and many of the faults were reactivated and record normal dip-slip, east side down, displacement.

Sampling Procedures and Analytical Methods

Samples were collected as either block samples or drilled cores orientated by magnetic and or Sun compasses. The Middle Jurassic La Negra Formation was sampled at 21 sites along two west to east traverses (Figure 2). Samples were collected mainly from porphyritic lavas (17 sites) of basaltic to andesitic composition, although devitrified ignimbrites and rhyolitic lavas were also sampled (Table 2). Structural control was obtained from orientation of adjacent sedimentary strata or primary igneous features consistent with more distant sedimentary strata.

Samples were also collected from dike swarms which intrude five Triassic to Early Cretaceous granitic plutons. Dikes were sampled from the Vetado, Flamenco, Las Animas, and Las Tazas plutons to the west of the AFZ, and from the Remolino pluton to the east (Figure 2). As noted previously, the dike swarms of the Flamenco (155 Ma), the Las Animas (153 Ma), and Las Tazas (129 Ma) plutons have been directly dated by *Dallmeyer et al.* [1996]. The youngest paleomagnetically sampled dike swarm is that which cuts the 126 Ma Remolino pluton [*Dallmeyer et al.*, 1996], and in common with the other Middle Jurassic-Early Cretaceous plutons, we believe that this swarm

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Table 2. Site Mean Directions From the La Negra Formation

Site	Lithology	n/N^o	Uncorrected		Tilt Correction	Corrected		k	α_{95}	Site Location	
			D, deg.	I, deg.		D, deg.	I, deg.			Longitude West	Latitude South
AT69[a]	ignimbrite	11/11	26.0	-41.7	254/21	44.8	-55.1	70.6	5.5	70°28'00"	26°08'30"
AT70	basaltic/andesitic lava	6/8	152.6	63.6	224/40	294.9	69.3	73.9	7.8	70°27'30"	26°08'15"
AT71	andesitic lava	6/9	291.1	-4.9	324/36	293.8	14.4	188.4	4.9	70°26'00"	26°08'30"
AT72[a]	basaltic/andesitic lava	8/9	234.4	7.1	338/24	227.7	30.6	40.0	8.9	70°25'37"	26°08'20"
AT74	andesitic lava	9/11	171.1	7.5	155/10	169.6	10.2	171.9	3.9	70°23'15"	26°08'15"
AT75[a]	andesitic lava	5/9	40.6	9.6	358/32	40.1	-12.2	13.3	21.8	70°23'03"	26°08'40"
AT76[a]	rhyolite lava	9/18	2.2	-48.3	206/52	60.9	-42.2	17.0	12.9	70°21'45"	26°07'30"
AT77	basaltic/andesitic lava	5/7	277.1	-48.0	233/13	265.6	-57.8	77.4	8.8	70°21'45"	26°07'30"
AT78[a]	basaltic lava	12/19	33.4	-37.2	251/16	45.5	-45.6	38.0	7.1	70°21'18"	26°07'30"
AT79[a]	andesitic lava	19/19	211.4	43.2	220/09	221.2	43.3	74.7	3.9	70°24'00"	26°10'20"
AT80[a]	basaltic lava	8/11	54.1	-42.2	192/23	31.6	-54.3	38.8	9.0	70°23'30"	26°10'30"
AT81	brecciated lava	7/7	24.4	-60.0	332/60	270.9	-50.9	32.6	10.7	70°37'00"	25°44'00"
AT82[a]	andesitic lava	8/9	241.1	1.5	352/30	237.9	29.3	43.9	8.6	70°36'00"	25°45'00"
AT83[a]	andesitic lava	7/9	225.5	18.8	320/30	223.5	48.7	22.3	13.6	70°35'00"	25°44'00"
AT84[a]	andesitic lava	6/7	25.9	-5.0	356/44	15.0	-24.0	17.2	16.9	70°35'00"	25°45'00"
AT85[a]	andesitic lava	7/7	231.7	-23.6	017/57	236.4	12.7	85.2	6.7	70°35'00"	25°45'00"
AT86[a]	basaltic/andesitic lava	4/6	25.6	16.5	328/42	25.3	-18.8	59.1	23.2	70°35'00"	25°43'50"
AT87	basaltic/andesitic lava	7/7	286.7	15.4	350/17	290.2	30.3	133	5.3	70°36'00"	25°47'00"
AT88[a]	andesitic lava	7/7	50.7	4.3	358/28	48.2	-18.1	34.6	10.4	70°36'00"	25°47'00"
AT89[a]	andesitic lava	7/7	43.4	-38.2	356/22	25.8	-52.2	76.2	7.3	70°36'00"	25°47'00"
Mean		14/20	40.2	-17.0				8.1	14.9		
						42.0	-35.5	17.9	9.6		

The notation is as follows: n/N^o, where n is the number of samples contributing to the site mean and n^o is the total number of samples in the site; D and I are the mean declination and inclination of the site mean magnetization vector; k and α_{95} are *Fisher's* [1953] statistical parameters.

[a] Sites included in the formation mean calculation.

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too was intruded shortly after emplacement of the host pluton. The oldest dike swarm sampled intrudes the Late Triassic Vetado pluton, but samples failed to yield a reliable $^{40}Ar/^{39}Ar$ date. The swarm is truncated by the Middle Jurassic Las Animas pluton and its dike swarm which is structurally distinct from that of the Vetado pluton, as shown by differing extension directions. The age of the dike swarm cutting the Vetado pluton is therefore bracketed as being pre-155 Ma (the age of the Las Animas pluton) and after the Triassic age of the host intrusion itself [*Berg and Breitkreuz*, 1983]. Given the compositional contrast between the host pluton and the dikes, we suggest that these dikes may have acted as feeders for the Early to Middle Jurassic La Negra Formation and are therefore most likely of the same age as the volcanic rocks as opposed to being related to a late stage of pluton emplacement itself in this case. The dikes are predominantly fine to medium grained and of basaltic-andesitic composition with the exception of two which are dacitic in composition. Although the paleohorizontal for the dikes swarms cannot be established unequivocally, our structural data, as noted previously, show that it is unlikely that the dikes have suffered any substantial postemplacement tilting.

Remanent magnetizations were measured using Molspin MS2 or Geofyzika JR4 spinner magnetometers and specimens were stepwise demagnetized by thermal or two-axes tumbling alternating field (AF) methods. Thermal demagnetization was used for over 85% of specimens as this method tended to provide better resolution of individual remanence components; however, at least one and normally more specimens from each site were demagnetized using AF methods to allow comparison of demagnetization behavior. Remanence directions were determined using principal component analysis [*Kirschvink*, 1980] or for a few (five) sites, by combined analysis of remagnetization circles and stable end points [*McFadden and McElhinny*, 1988]. Mean directions for each geological unit were calculated by combining site mean directions [*Fisher*, 1953]. Magnetic carriers were identified by a combination of demagnetization behavior, acquisition of isothermal remanent magnetization (IRM) and demagnetization of three component IRMs [*Lowrie*, 1990] using fields of 800 mT, 300 mT, and 50 mT. This method allows the identification of the magnetic carriers through correlation of the unblocking temperatures and magnetic coercivities during stepwise thermal demagnetization.

Paleomagnetic Results

Sample behavior during demagnetization in both the volcanic rocks and the dikes is variable, but a distinctive, well-grouped, characteristic remanence (ChRM) was recovered from most sites. This ChRM direction is in the northeast quadrant with negative inclination or its antipode, after tilt correction of the La Negra Formation and in situ for the dikes. Most samples exhibit a multicomponent natural remanent magnetization (NRM) behavior with the ChRM usually being identified as a linear segment in the demagnetization trajectories between 200°C and 580°C (Figures 4a, 4c and 4e), or typically between 20 and 100 mT (Figure 4b) or rarely from high temperature endpoints (Figure 4d) and remagnetization circles (Figure 4f). In multicomponent samples, most have a low unblocking temperature/low-coercivity component which normally exhibits a consistent within site direction. These low unblocking components fall into three categories defined on the basis of their directions and sample NRM intensities: (1) those from samples with total NRM intensities in the normal range for these rocks and whose directions lie close to the present geomagnetic field direction, which are interpreted as present-day VRMs (Figures 4a, 4c and 4e); (2) those from samples with very high total NRM intensities which rapidly reduce on AF demagnetization and whose directions are different from both the ChRM and the present-day geomagnetic field direction; these we interpret as the effects of lightning strikes; and (3) those which exhibit normal total NRM intensities and yield consistent within site directions but which are not directed toward the present-day geomagnetic field direction (Figure 4d). At these sites the ChRM is recovered at higher demagnetization levels, and the samples do not behave in a viscous manner. These components may be an ancient thermochemical component related to alteration; however, the mean site directions do not form a coherent group, and we cannot therefore interpret them reliably.

Demagnetization behavior and IRM acquisition/demagnetization experiments indicate that magnetite is the dominant magnetic carrier in most sites (Figure 5). As shown in Figure 5a most samples are dominated by a low-coercivity fraction which saturates at about 300 mT. The major decrease in intensity during demagnetization of the low-coercivity component, between 100°C and 300°C, is interpreted as an effect of multi-domain magnetite. The decrease in magnetization between 300°C and 570°C possibly reflects a range of grain sizes. Less commonly there is a mixture of carriers (Figure 5b). In such samples there is a rapid rise in the intensity of the IRM acquired up to 300 mT, but the samples do not saturate in the maximum applied field of 800 mT. This behavior, coupled with the unblocking temperatures of 670°C, indicates a hematite carrier. Demagnetization data reveal that hematite has the same direction as that carried by the magnetite (Figure 4c). Hematite has been identified as a significant carrier at sites AT70 and AT88 in the La Negra Formation and in three dike sites from the Flamenco pluton.

Of 21 sites analyzed from the La Negra Formation, 20 yield a well-grouped within-site direction of magnetization (Table 2 and Figure 6). Of these, 14 carry the ChRM and pass a fold test at the 95% confidence level [*McElhinny*, 1964]; furthermore, the tilt correction results in steepening of the inclination of the formation mean from -17.0° to -35.5°, which is much closer to the expected -44.6° inclination for a Middle Jurassic remanence [*Beck*, 1988]. These 14 sites have both normal and reversed polarities which are antipodal at the 95% confidence level (C classification of *McFadden and McElhinny* [1990]). These tests suggest that the magnetization is primary or was acquired soon after formation and prior to deformation. Of the six rejected sites, five show a direction unrelated to the ChRM and may be recording an intermediate polarity direction. The sixth is from a brecciated lava which has a pretilt direction close to the tilt-corrected formation mean and may have been remagnetized after tilting, perhaps becoming preferentially remagnetized by fluid flow due to its high porosity.

All five dike swarms, comprising 49 sites, possess mean directions which are indistinguishable from the mean direction of the La Negra Formation at the 95% confidence level (Table 3 and Figure 6). Four of the five dike swarms show mixed polarity, which again suggests that the magnetization was primary or acquired shortly after emplacement. All dikes within the youngest (post-126 Ma) Remolino swarm have normal polarity which is consistent with their emplacement having taken place during the long mid-Cretaceous normal polarity period. A total of 13 sites have been rejected from the calculations of the mean

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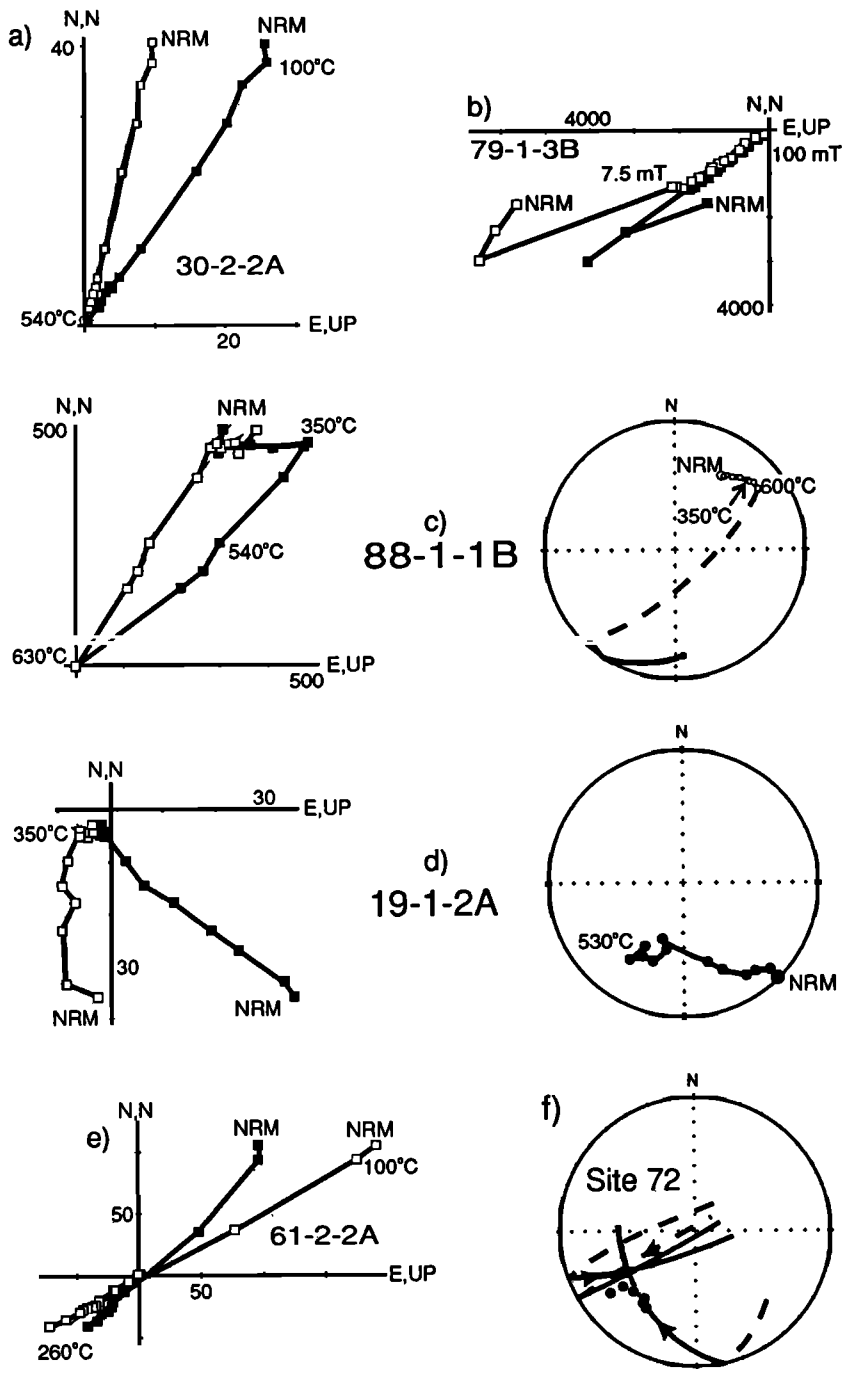


Figure 4. Orthogonal and equal-area stereonet projections of sample demagnetization behavior for typical samples (Figures 4a-4e), and an example of combined endpoint and remagnetization circle analysis for a single site (Figure 4f). In the orthogonal diagrams, open and solid symbols are the vertical and horizontal projections, respectively, magnetization intensities are in mA m^{-1}. On the stereonet projections, open and solid symbols are negative and positive inclinations, respectively.

directions because of either (1) high within site dispersion (four sites); (2) stable well-defined directions of magnetization but which are significantly removed from the ChRM which may, for example, record intermediate directions acquired during a polarity transition (six sites); or (3) a group of three sites from a small sampling area in the Vetado pluton dike swarm which have declinations approximately 20° lower than the group mean. These sites lie close to a large fault [*Mercado*, 1978] and may have been selectively remagnetized during late fault movements. The agreement of remanence directions between the plutons and the tilt-corrected result from the La Negra Formation accords with the structural evidence that the dikes have not suffered substantial post emplacement tilting.

The volcanic rocks of the La Negra Formation and the dikes all show varying degrees of low-temperature alteration and cataclasis in proximity to major faults. The alteration is characterized by epidote pseudomorph replacement of ferromagnesian minerals, sericitization of feldspars, and, in some cases, oxida-

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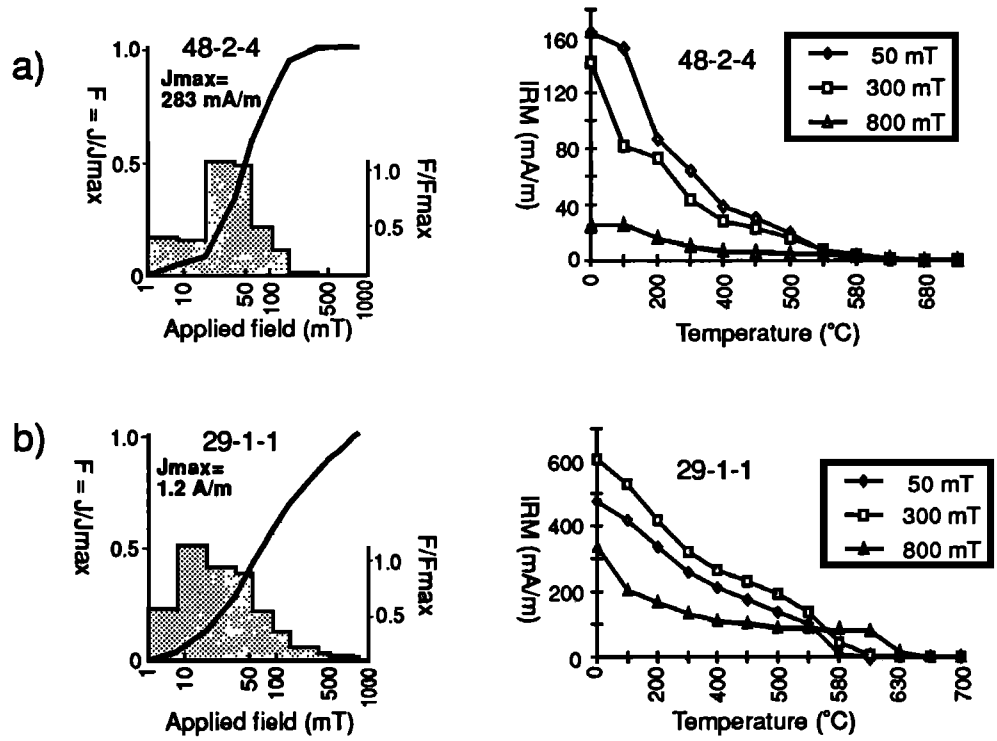


Figure 5. Examples of IRM acquisition and demagnetization curves [*Lowrie*, 1990] indicative of (a) a sample with a dominant magnetite carrier and (b) both magnetite and hematite carriers.

tion. The cause of this alteration is not clear, as at least three contributing factors may be responsible. Deuteric alteration may have occurred, especially in the volcanic rocks of the La Negra Formation, where the intercalated marine sedimentary horizons would have retained fluids and the subsequent extrusion of further lavas would provided heat for the alteration. Deuteric alteration may also have affected the dikes, particularly if the host plutons were still warm during dike emplacement. With this type of alteration the magnetization in the rocks will not be primary as such but is likely to have been acquired soon after emplacement as a result of thermochemical processes. Such early alteration may be responsible for the demagnetization behavior observed in some samples which record both normal and reversed components which are antipodal to each other (Figures 4b and 4f). The second type of alteration affecting some of these rocks is the result of action by low-temperature mineralizing fluids along fault systems or in some cases the plutons themselves. Wherever possible sampling was conducted in areas which were not obviously affected by alteration. If this type of alteration has occurred in sampled sites, the magnetization recorded in the rocks may not be primary and could be significantly younger than the age of the extrusion or emplacement. Third, low-temperature subgreenschist facies metamorphism has been reported from similar volcanic suites to the south of this study area [*Aguirre et al.*, 1989] and is thought to be the result of burial, probably occurring rapidly after extrusion or emplacement. We have not, however, observed evidence of this in our samples and hence suspect it is unlikely to have substantially affected the remanences. Although we recognize the pervasive nature of this alteration we believe, because of the positive fold and reversal tests, that the magnetizations of all the sampled units were acquired either as primary TRMs associated with emplacement or early thermochemical remanences as a result of pervasive deuteric alteration which occurred shortly after emplacement.

Interpretation and Discussion of the Paleomagnetic Results

Comparison With Reference Directions from the Stable Craton

The apparent polar wander path and mean paleomagnetic reference poles for South America during the Jurassic and Cretaceous have been the subject of some debate as several of the contributing poles are poorly determined. The most recent contribution to defining the mean poles for this period is that of *Raposo and Ernesto* [1995], who report high-quality poles for the Early and mid-Cretaceous, but reliable Jurassic reference poles remain a problem. For ease of comparison with previously published results we have used the reference poles of *Beck* [1988], although noting that comparison of our directions with more recent reference poles [e.g., *Roperch and Carlier*, 1992; *Raposo and Ernesto*, 1995] does not significantly alter the magnitude of the tectonic rotations or their errors.

Comparison of the observed inclinations with the expected inclinations derived from the reference poles from four of the dike swarms (Vetado, Flamenco, Las Animas, and Las Tazas) shows no statistically significant latitudinal transport, although the results from the La Negra Formation and the Remolino dike swarm show southward displacement of the area since the time of remanence acquisition. Such latitudinal displacement would imply southward displacement of the forearc sliver or terrane, consistent with our own tectonic interpretation, but we cannot reconcile this with the four results which do not show significant latitudinal transport. Interpretation of previous results from the Coastal Cordillera shows that they do not demonstrate a consistent sense of latitudinal transport [e.g., *Beck et al.*, 1994], although we note a tendency of the results from the Coastal Cordillera of northern Chile to suggest southward transport. In comparison to other reference poles [*Roperch and Carlier*, 1992; *Raposo and Ernesto*, 1995] the pattern of latitu-

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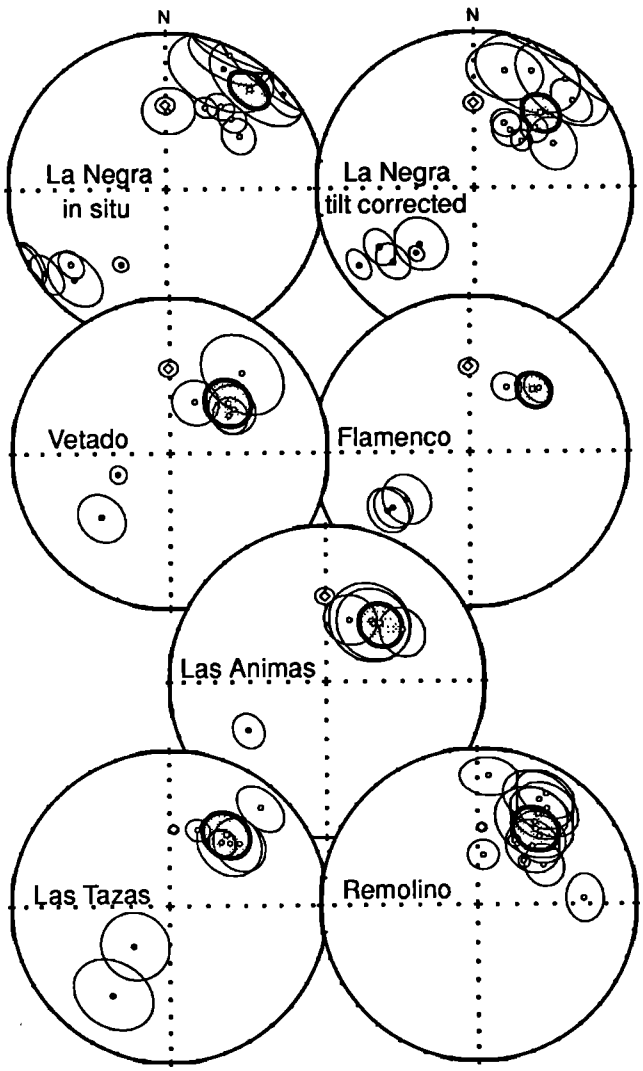


Figure 6. Equal-area stereonet projections of site mean directions and 95% confidence circles for all units in Tables 2 and 3. The mean direction for each unit is shown with its 95% confidence circle shaded. Open and solid symbols represent negative and positive inclinations, respectively. The results for the La Negra Formation are shown both in situ and after tilt correction. The expected directions calculated from the reference poles of *Beck* [1988] are shown as open diamonds with their given 95% error circles.

dinal transport is also inconsistent; hence we must assume that the prime reason for such inconsistencies is still the lack of sufficiently accurate reference poles.

Comparison of the observed and expected declinations indicates clockwise rotations ranging from 34° to 50° with a mean of 42° which lies within the errors associated with all the units sampled (Table 4). It is notable however that the data from Jurassic rocks have an average of 10° more rotation than the Early Cretaceous Las Tazas and Remolino dike swarms. This may indicate that rotation was first initiated at the earliest phase (132 Ma) of ductile strike-slip fault displacement on the AFZ itself as is also suggested by our structural studies. Although this discrepancy is found regardless of the reference poles used for the calculations, we consider the overlap of errors such that we cannot yet reliably discern two stages of rotation without further studies to confirm this. We therefore interpret the data conservatively as representing an average 42° clockwise rotation of this part of the Coastal Cordillera and that the variation may represent slight differences in total rotation of individual structural blocks within a single structural domain. Any model advanced to explain the data must also account for the fact that the Early Cretaceous Remolino and Las Tazas dike swarms, immediately east and west of the Atacama Fault Zone, yield nearly identical results. To the north of our study area, in the Paposo segment of the Atacama Fault Zone, *Forsythe and Chisholm* [1994] have also shown that clockwise rotations exist to both east and west of the AFZ, although in this case the limited Cretaceous data from east of the fault zone show greater rotation than the Jurassic data to the west.

Large-Scale Models of Deformation

Paleomagnetic studies of Mesozoic and Tertiary rocks from the Andean margin, as previously noted, show a change from counterclockwise rotations north of the bend to clockwise south of it (Figures 1a and 1b). This has led several workers to advocate large-scale tectonic mechanisms to explain the results. We outline these models below and summarize their applicability to our results.

Oroclinal bending [*Carey*, 1955]. In this model, an originally straight Andean margin was bent by passive rotation of the crust to the north and south of Arica to form the Deflection (Figure 7a). Geological arguments against such a simplistic model of large-scale rigid body rotation include the lack of obvious extensional features on the inboard side of the of the deforming block (see *Beck* [1988] for discussion).

Differential shortening [*Isacks*, 1988]. In this modification of oroclinal bending, the northern and southern limbs of the orocline are passively rotated as a consequence of differential shortening across a preexisting bend in the Andean margin. This would have enhanced the bend, while shortening would account in part for the significant uplift of the region (Figure 7b). In the original model, *Isacks* [1988] predicted a maximum of 10° of clockwise rotation for this part of the Coastal Cordillera of northern Chile. The most recent reviews of the paleomagnetic database for the region have been used to provide support for this hypotheses [e.g., *MacFadden et al.*, 1995; *Butler et al.*, 1995] at least for data of Late Tertiary age (Figure 1a). Some workers, however, have suggested that differential shortening could account for much larger rotations in northern Chile but without detailing the reasons for this [e.g., *Watts et al.*, 1995]. In contrast to these views, some workers interpret the variability of results from pre-Tertiary rocks (Figure 1b), from northern Chile and southern Bolivia, to indicate that the differential shortening hypotheses is invalid [e.g., *Forsythe and Chisholm*, 1994; *Somoza*, 1994]. We would argue that the amount of rotation in this study clearly exceeds the predictions of the original differential shortening model [*Isacks*, 1988], and hence this mechanism cannot provide an adequate explanation for the bulk of the observed rotation. We cannot, however, rule out, categorically, that a small component of the rotation may be caused by such a mechanism acting at the very large scale in conjunction with more localized deformation; thus the observed rotations may, in fact, be compound as has also been suggested by others [e.g., *Riley et al.*, 1993; *Beck et al.*, 1994; *Butler et al.*, 1995].

Localized Block Rotations

The pattern of rotations from Mesozoic and older rocks of the Chilean Coastal Cordillera is complex (Table 2 and Figure 1b), most greatly exceed the 10° predicted by the *Isacks* [1988]

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Table 3. Site Mean Directions From the Dike Swarms

Site	n/N^o	D, deg	I, deg	k	α_{95}	Site location	
						Longitude West	Latatitude South
				Vetado			
AT1	4/6	18.0	-34.7	24.2	19.0	70°29'30"	26°22'30"
AT2	4/6	56.5	-45.8	11.7	29.6	70°29'30"	26°22'30"
AT3[a]	5/6	58.0	-52.8	69.0	9.3	70°29'40"	26°21'35"
AT4[a]	6/6	42.4	-31.3	12.7	20.8	70°29'45"	26°21'45"
AT5	4/6	12.4	-58.2	834.4	3.4	70°29'45"	26°21'45"
AT6	6/6	18.2	-41.2	13.6	18.8	70°30'00"	26°21'50"
AT7[a]	5/6	247.6	60.6	585.6	4.6	70°30'00"	26°21'50"
AT8[a]	6/6	54.8	-49.9	37.0	12.4	70°30'00"	26°21'50"
AT10[a]	6/6	26.5	-60.2	37.4	11.9	70°29'00"	26°21'45"
AT11[a]	6/6	226.4	39.0	32.9	12.3	70°29'00"	26°21'45"
Mean	6/10	48.9	-49.6	31.9	12.1		
				Flamenco			
AT61[a]	7/7	233.6	39.1	35.9	10.2	70°39'10"	26°33'50"
AT62[a]	7/7	232.5	47.9	23.9	12.7	70°39'10"	26°33'50"
AT63	7/7	350.4	-67.8	39.5	9.8	70°39'45"	26°34'10"
AT64[a]	7/7	233.9	35.9	30.4	11.1	70°39'45"	26°34'15"
AT66[a]	7/7	29.0	-50.6	68.4	7.4	70°39'00"	26°34'10"
AT67[a]	7/7	46.8	-40.2	67.6	7.4	70°38'45"	26°34'05"
Mean	5/6	45.6	-43.0	76.3	8.8		
				Las Animas			
AT17	3/9	229.6	38.0	29.3	23.2	70°25'50"	26°19'50"
AT18[a]	4/4	237.9	40.9	130.6	8.6	70°25'50"	26°19'50"
AT19[a]	7/9	57.1	-41.7	25.4	12.2	70°26'00"	26°20'00"
AT20[a]	7/7	21.8	-55.1	22.1	13.3	70°26'00"	26°20'35"
AT21[a]	7/7	36.3	-50.8	9.5	20.7	70°25'50"	26°21'00"
AT22[a]	7/8	39.4	-50.6	11.5	18.7	70°26'00"	26°21'10"
Mean	5/6	44.0	-48.6	47.2	11.2		
				Las Tazas			
AT27	7/7	193.0	4.7	279.3	3.6	70°21'30"	26°24'00"
AT28[a]	6/8	43.9	-43.8	18.4	16.6	70°21'30"	26°24'00"
AT29[a]	7/7	39.8	-46.5	75.5	7.0	70°21'10"	26°23'30"
AT30[a]	5/7	43.0	-15.8	51.7	11.5	70°21'00"	26°23'10"
AT31	6/6	154.8	28.5	21.9	10.5	70°21'35"	26°30'45"
AT32[a]	6/8	20.4	-47.0	155.9	6.0	70°21'35"	26°30'45"
AT33[a]	6/6	220.9	62.1	17.3	17.7	70°21'30"	26°31'00"
AT34[a]	6/6	212.7	32.5	17.8	19.2	70°21'30"	26°31'00"
AT37	6/6	155.8	44.7	333.7	3.7	70°23'40"	26°39'00"
AT38[a]	5/6	48.9	-40.2	34.5	13.6	70°21'00"	26°35'00"
Mean	7/10	38.7	-41.5	26.4	12.0		
				Remolino			
AT39[a]	4/7	4.9	-18.5	57.8	12.2	70°18'00"	26°47'30"
AT40[a]	6/7	30.0	-23.3	17.8	16.3	70°18'00"	26°47'30"
AT41[a]	7/7	36.3	-25.5	17.9	14.7	70°17'50"	26°47'30"
AT42[a]	6/7	59.5	-49.6	40.6	10.6	70°17'50"	26°47'30"
AT43[a]	7/7	29.8	-51.5	347.5	3.2	70°17'45"	26°47'30"
AT44	4/7	260.0	-28.1	999.9	1.8	70°16'00"	26°47'32"
AT45[a]	8/18	33.7	-30.6	17.2	14.0	70°17'30"	26°44'45"
AT46[a]	9/17	34.5	-36.0	8.2	19.2	70°17'30"	26°44'45"
AT47[a]	10/10	47.3	-57.3	268.2	3.0	70°17'30"	26°44'45"
AT48[a]	15/17	42.3	-41.6	9.0	13.5	70°17'30"	26°44'45"
AT49[a]	14/14	6.2	-64.0	26.5	7.9	70°17'30"	26°44'45"
AT50	15/15	119.2	27.8	107.9	3.7	70°17'30"	26°44'45"
AT53[a]	5/7	86.8	-32.6	49.1	11.0	70°16'10"	26°35'45"
AT54	7/7	232.3	-22.8	49.0	8.7	70°16'10"	26°35'45"
AT55[a]	3/7	31.5	-18.3	88.2	13.2	70°16'10"	26°35'45"
AT56[a]	4/6	43.4	-47.8	45.2	13.8	70°15'50"	26°31'40"
AT57	3/7	48.3	-30.0	23.7	25.7	70°15'50"	26°31'40"
Mean	13/17	37.2	-39.3	13.8	11.6		

Notation as in Table 2.
[a] Sites included in the formation mean calculation.

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Table 4. Calculation of Rotation and Flattening With Respect to Reference Directions of *Beck* [1988]

Formation	Observed			Reference			Rotation, deg	Flattening, deg
	D, deg	I, deg	α_{95}	D, deg	I, deg	A_{95}		
La Negra	42.0	-35.5	9.6	359.0	-44.6	4.6	43.0±10.8	-9.1±8.5
Vetado	48.9	-49.6	12.1	359.0	-44.6	4.6	49.9±16.0	5.0±10.4
Flamenco	45.6	-43.0	8.8	359.0	-44.6	4.6	46.6±11.0	-1.6±7.9
Las Animas	44.0	-48.6	11.2	359.0	-44.6	4.6	45.0±14.6	4.0±9.7
Las Tazas	38.7	-41.5	12.0	3.1	-49.2	2.4	35.6±13.2	-7.7±9.8
Remolino	37.2	-39.3	11.6	3.1	-49.2	2.4	34.1±12.4	-9.9±9.5

Calculated parameters after *Beck* [1988] and *Demarest* [1983].

model [e.g., *Riley et al.*, 1993; *Forsythe and Chisholm*, 1994; this study], although some areas exhibit only small or no significant rotation [*Irwin et al.*, 1987; *Jesinkey et al.*, 1987]. In the Mejillones Peninsula, just north of Antofagasta in northern Chile, rotations appear to be very localized and predominantly counterclockwise [*Hartley et al.*, 1992b]. The data set as a whole is interpreted by ourselves and others [e.g., *Forsythe and Chisholm*, 1994; *Somoza*, 1994] to indicate that block rotation about local vertical axes must be of greater significance than large-scale mechanisms in producing the overall variation in the observed magnitudes of rotations. Proposed mechanisms of localized block rotations which may have affected the Chilean Coastal Cordillera include the following:

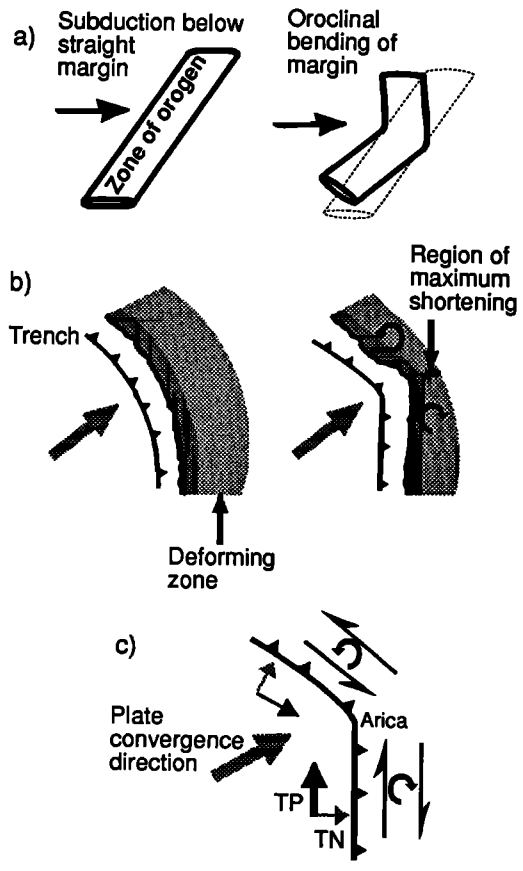


Figure 7. Cartoons of large-scale mechanisms proposed to explain the observed rotations about the Arica Deflection. (a) Oroclinal bending of an originally straight margin [*Carey*, 1955], (b) differential shortening in the interior of the margin [*Isacks*, 1988], and(c) oblique convergence at a preexisting bend leading to localized but systematic rotations on either side of the deflection driven by regional shear [*Beck*, 1987, 1988].

Distributed crustal shear [*Beck*, 1987, 1988]. This model assumes that the Arica Deflection is an original feature of the Andean margin and that convergence across this bend generates in situ block rotation as a consequence of a component of trench parallel shear. In this case, small structurally controlled blocks are rotated clockwise south of the deflection in a right-lateral shear zone and conversely counterclockwise north of the deflection (Figure 7c). Although such a model might explain the sense of rotations, it should lead to a decrease in rotation with distance from the trench [*Hartley et al.*, 1988] which is not seen in the collective data set for the entire region (Table 1 and Figures 1a and 1b). Furthermore, it would require that the major fault systems of northern Chile should be dominantly right-lateral in their displacement when, in fact, they are dominantly left-lateral [e.g., *Brown et al.*, 1993; *Tomlinson et al.*, 1993; *Arevalo*, 1994, 1995]. We therefore cannot support this model advocating right-lateral crustal shear as being responsible for the observed clockwise rotations on the basis that left-lateral shear is dominant in this part of the Andean margin.

Oblique convergence coupled with subduction rollback [*Hartley et al.*, 1988]. In this model, block rotation is a consequence of transtension in the forearc sliver. We have considered such a mechanism as being plausible given the long time, Late Triassic to Early Cretaceous, when the margin was clearly part of a retreating subduction margin dominated by extension. Our paleomagnetic data, however, clearly indicate that the bulk, if not all, of the rotation postdates the emplacement of the Remolino pluton at 126 Ma. After this time, evidence of extension in this part of the arc is limited, but such a model would require that large volumes of magma should be emplaced as plutons and/or sedimentary basins should develop in the extensional zones of the overall transtensional system to accommodate the rotating blocks.

Left-lateral shear on the Atacama Fault Zone [*Forsythe and Chisholm*, 1994]. In this model, left-lateral displacement on the AFZ and its subsidiary NW trending faults leads to a domino style block rotation. *Forsythe and Chisholm* [1994] estimate a rotation value based on simple structural restoration which is consistent with their observed paleomagnetic rotation. This restoration assumes an average apparent horizontal fault displacement on the NW trending strike-slip faults, relative to a fixed AFZ. Similar calculations based on selected parts of our field area also yield structurally defined rotations which are consistent with the paleomagnetic rotations. We regard such geometrical restorations with some caution, as our observations indicate that the NW faults rarely have a pure strike-slip geometry or a single phase of displacement; hence the horizontal offset is both apparent and compound.

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Our Model

We consider two possible variants of left-lateral strike-slip faulting leading to clockwise rotation, they differ primarily in the emphasis given to the role of the Atacama Fault Zone.

Model A: The Atacama Fault Zone. In this model (Figure 8a) the AFZ is interpreted as the surface expression of a single deep-seated crustal fault whereby the NW trending left-lateral faults are synthetic and the much rarer NE trending right-lateral faults are antithetic. Such a model is consistent with both theoretical and observed examples [*Naylor et al.*, 1986] and could lead to the observed fracture pattern (Figure 3). In this model we might expect both observed rotations and fault displacements on the NW faults to increase toward the principal fault. In contrast to this we tend to find that displacement on the NW trending faults commonly increases away from the AFZ toward the coast and that the youngest plutons adjacent to the principal fault itself show the least rotation (i.e., the results from the Las Tazas and Remolino plutons). Furthermore, at the largest scale, the AFZ is cut and significantly displaced by the largest NW trending faults, indicating that movement on these faults was long-lived and to a large extent postdates the principal movements of the AFZ itself (Figure 3).

Model B: A crustal scale left-lateral shear zone. This model is mechanically identical to that of *Forsythe and Chisholm* [1994] but differs in that we do not consider the AFZ to be the principal eastern boundary of the rotated blocks. We propose that the Central Valley Shear Zone acted as this boundary in mid-Late Cretaceous times. The Central Valley Shear Zone would have acted as the main trench linked strike-slip fault defining a broader forearc sliver than had previously been the case. During this period the NW trending faults, initiated by the AFZ, would have been reactivated and extended and hence would have cut and displaced the AFZ itself. These faults extend eastward to the Central Valley Shear Zone and form an integral part of the mid-Late Cretaceous Atacama Fault System. The rotations we believe occurred in the same domino style manner as proposed by *Forsythe and Chisholm* [1994] for the Paposo segment of the AFZ and previously advocated by others for similar systems of left-lateral faults [e.g., *Ron et al.* 1984; *Nur et al.*, 1988; *Schreurs*, 1994, and references therein]. In such a model, clockwise rotations are generated by left-lateral transpression across the shear zone (Figure 8b). The blocks are defined by the major NW trending oblique-slip faults which have a small component of dip-slip, NE side down displacement in addition to the left-lateral component seen in map view. This dip-slip displacement accommodates some of the change of shape of the blocks during rotation and extension of the shear zone due to E-W or NW-SE shortening.

Given geochronologic data on the age of displacements along the AFZ, model A would require that the bulk of rotation was Early Cretaceous in age. Although such motion may account for the 10° or so difference in rotations observed between Jurassic and Cretaceous rocks, should this be discrepancy be proven by further investigations, we favor model B as the principal cause of the rotations because it better explains the observed geology across the entire Coastal Cordillera. In particular, it explains offsets in the AFZ (up to 70 km) [*Brown et al.*, 1993], the clockwise rotation of offset segments of the AFZ (for example, near the city of Copiapo at 27°S segments of the AFZ trend at 040 to 050 [*Arevalo*, 1995]) and the rotations found on both sides of the AFZ [e.g., this study; *Forsythe and Chisholm*, 1994]. Furthermore, the outcrop pattern of the Early to Middle Cretaceous sedimentary rocks in the east of the Coastal Cordillera suggests to us that these are preserved in extensional zones created by the blocks rotating away from the Central Valley Shear Zone (G.K. Taylor et al., manuscript in preparation, 1996). Such accommodation structures are to be expected in a

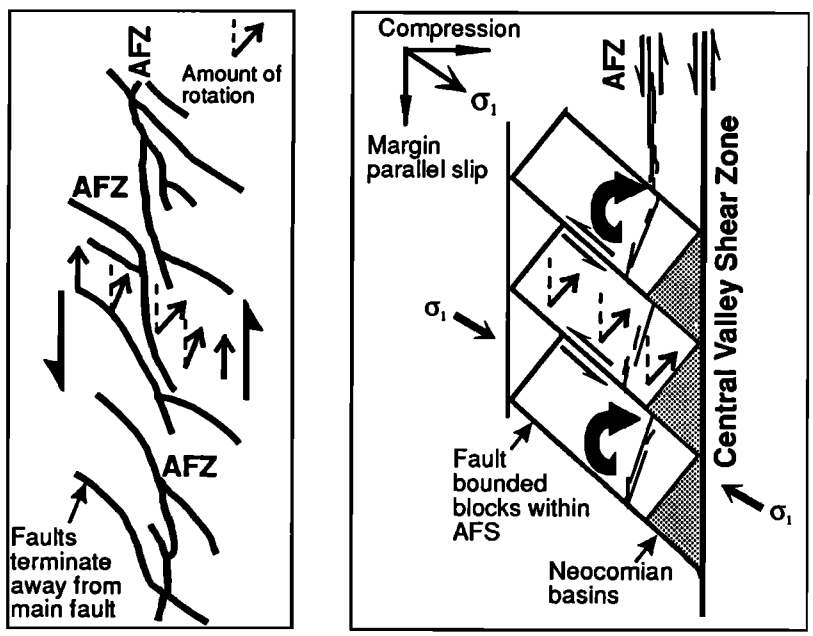


Figure 8. Proposed models for the observed rotations of this study. In model A, rotation is caused by strike-slip motion on the AFZ alone, while in model B, our preferred solution, the AFZ is not responsible for the rotations but is dissected and rotated within a broader crustal scale shear zone whose eastern boundary is marked by the Central Valley Shear Zone.

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domino-type system of fault block rotations. Although the cartoon of this model (Figure 8b) implies that these are rigid blocks unaffected by internal deformation, we believe that the blocks themselves deform to accommodate some change of shape during rotation. This deformation most obviously manifests itself in the repeated reactivation of the NW trending faults as well as minor folds and thrusts.

This model of block rotation might imply that oblique convergence with S-SE directed subduction beneath the Andean margin was responsible for the observed left-lateral strike-slip shear. While this accords with the suggested subduction direction for mid-Late Cretaceous times [*Duncan and Hargreaves*, 1984; *Thiele and Pincheira*, 1987], little direct evidence of this exists, and extrapolation from the deformation of the overriding plate is, as of yet, not possible. In particular, we note the dominance of left-lateral strike-slip fault systems in Tertiary time when the subduction direction is known to be east to northeast [*Tomlinson et al.*, 1993]. We are attempting to confirm the existence of our proposed crustal scale left-lateral shear zone with systematic structural and paleomagnetic studies to the east and west of the interpreted Central Valley Shear Zone.

Conclusions

The Jurassic La Negra Formation and Jurassic to Early Cretaceous dike swarms of this part of the Coastal Cordillera carry a ChRM whose characteristics are consistent with it having been acquired as either a primary TRM or early thermochemical remanence. Comparison with reference poles for the stable South American craton indicates that the region has suffered an average of 42° of clockwise rotation since 126 Ma, while latitudinal transport cannot yet be clearly identified due to doubt about the accuracy of the reference poles. Paleomagnetic directions from immediately to the west and east of the AFZ are identical, and hence we do not believe that the AFZ is the principal fault controlling the rotation. We propose that rotation was by domino-type rotation of blocks bounded by NW trending left-lateral faults which sole out into a newly recognized fault zone, the Central Valley Shear Zone which separates the Coastal Cordillera from the Central Valley of northern Chile. The timing of this rotation is believed to have been mid-Late Cretaceous, coincident with the abandonment of a Middle Jurassic to Early Cretaceous magmatic arc and its subsequent eastward migration. We believe this event to be linked to a major change in the tectonic setting of the margin as evidenced by the first phase of compressional tectonics recognized along the length of the Andean margin and marking the onset of the Andean orogenic cycle.

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